

Divisibility: Let a and b be two integers. a is said to divide b if there exists an integer c such that $b = ac$. Otherwise, we say that a doesn't divide b .

Ex: $a = 3$ $b = 6$.

$$6 = 3 \times 2 \quad \text{So, } 3 \text{ divides } 6.$$

It is written $3 \mid 6$.

3 doesn't divide 1 .

So $3 \nmid 1$.

" a divides b " is same as saying the following:

- (1) a is a factor of b .
- (2) b is a multiple of a .
- (3) a is a divisor of b .

Theorem (1): Let a be a non-zero natural number. Then the following statements are true.

(1) $a \mid 0$.

(2) $a \mid a$.

(3) If $a \mid b$ then $a \mid bc$ for all $b, c \in \mathbb{N}$

where $\mathbb{N}$ denotes the set of all natural numbers.

④ If $a|b$ & $b|c$ then $a|c$ $\forall a, b, c \in \mathbb{N}$

⑤ If $a|b$ & $a|c$ then $a|(b \pm c)$ $\forall a, b, c \in \mathbb{N}$

⑥ If $a|b$ & $b|a$ then $a=b$ $\forall a, b \in \mathbb{N}$.

Proof: ① $a \cdot 0 = 0 \Rightarrow a|0$.

② $a \cdot 1 = a \Rightarrow a|a$

③ $a|b \Rightarrow \exists$ an integer x such that

$$b = ax.$$

$$\Rightarrow bc = a(xc)$$

$$\Rightarrow a|bc.$$

④ $a|b \Rightarrow \exists$ an x such that $b = ax$

$b|c \Rightarrow \exists$ an y " " $c = by$

$$\text{So } c = by = a(by)$$

$$\Rightarrow a|c.$$

⑤ $a|b \Rightarrow b = ax$ for some x

$a|c \Rightarrow c = ay$ for some y

$$b \pm c = a(x \pm y)$$

$$\Rightarrow a|(b \pm c)$$

$$\textcircled{6} \quad a \mid b \Rightarrow b = ax \quad \text{for some } x.$$

$$b \mid a \Rightarrow a = by \quad \text{for some } y.$$

$$\text{so } a = by = axy.$$

$$\Rightarrow xy = 1.$$

$$\text{Since } a, b \in \mathbb{N}, \quad x, y \in \mathbb{N}.$$

$$\Rightarrow x = y = 1, \quad \Rightarrow a = b.$$

Well-ordering principle: Every non-empty subset of positive (or non-negative) integers has a smallest element.

Let S be the set of all even numbers.

then 2 is the smallest element in S .

Division Algorithm: Let $a, b \in \mathbb{N}$. Then there exists unique integers q and r such that

$$a = bq + r \quad \text{where } 0 \leq r < b.$$

Proof: Consider the set $S = \{a - bx \mid x \in \mathbb{N} \cup \{0\} \text{ and } a - bx \geq 0\}.$

$$\text{Then } S \subseteq \mathbb{N} \cup \{0\}$$

Claim: $S \neq \emptyset$

take $x = 0$ then $a - bx = a \in S.$

So by well-ordering principle S has a smallest element.

Let r be smallest element of S .

$$\text{and } r = a - bq \quad \text{--- (1)}$$

By the construction of S , we have $r \geq 0$.

If possible let $r \geq b$.

then $r = b + \alpha$ for some $\alpha \in \mathbb{N} \cup \{0\}$.

So from (1) we have

$$r = b + \alpha = a - bq$$

$$\Rightarrow \alpha = \cancel{a - b(q+1)} = a - b(q+1) \in \mathbb{N}$$

Since $\alpha \in \mathbb{N} \cup \{0\}$, $\alpha \in S$.

This contradicts the fact that r is the smallest element in S .

Hence $r < b$.

So we have proved the existence of q and r .

It remains to show that q and r are unique.

~~If possible~~ Let there exist q' & r' such that

$$a = bq' + r' \quad \text{where } 0 \leq r' < b. \quad (\text{Assume } q' \geq q)$$

$$\Rightarrow bq' + r' = bq + r$$

$$\Rightarrow r - r' = b(q' - q)$$

$\Rightarrow r - r'$ is a multiple of b .

Since $0 \leq r < b$ & $0 \leq r' < b$,

we have $r - r' \geq 0 \Rightarrow r = r' \Rightarrow q = q'$

Greatest Common Divisor (GCD): Let $a, b \in \mathbb{N}$.

A positive integer d is called a gcd of a and b if it satisfies the following two conditions.

- ① $d \mid a$ and $d \mid b$.
- ② If $c \in \mathbb{N}$ is such that $c \mid a$ & $c \mid b$, then $c \mid d$.

Ex: gcd of 6 and 10 is 2.

Theorem: Each pair of natural numbers has a gcd.

Proof: Let $a, b \in \mathbb{N}$.

Consider $X = \{ax + by \in \mathbb{N} \mid x, y \in \mathbb{Z} \text{ (set of integers)}\}$

$$X \subseteq \mathbb{N}.$$

Clearly $X \neq \emptyset$.

So, by well ordering property, X has a smallest element, say d .

We claim d is the gcd of a and b .

Since d is the smallest positive integer in X ,

$$d = au + bv \text{ for some } u, v \in \mathbb{Z}.$$

Need to show

① $d \mid a$ and $d \mid b$

② If $c \mid a$ & $c \mid b$ then $c \mid d$.

Proof of (ii)

$$\text{if } c|a \text{ \& } c|b$$

$$\text{then } c|au+bv \Rightarrow c|d.$$

To prove (i), we will first show that $d|a$.

By division algorithm, $\exists$ unique integers q and r such that

$$a = dq + r \quad \text{where } 0 \leq r < d.$$

Claim: $r = 0$.

If $r \neq 0$, then $0 < r < d$.

$$\begin{aligned} r &= a - dq = a - (au + bv)q \\ &= a(1 - uq) + b(-vq) \in X. \end{aligned}$$

$$\Rightarrow r \in X \text{ \& } r < d.$$

This contradicts the fact that d is the smallest element of X .

$$\Rightarrow r = 0 \Rightarrow a = dq \Rightarrow d|a.$$

In a similar manner, we can prove $d|b$.

So, d is the gcd of a and b .

Corollary! Let $a, b \in \mathbb{N}$. Then $\gcd(a, b) = ax + by$
for some $x, y \in \mathbb{Z}$.

Remark: For any two natural number, gcd is unique.

Let $a, b \in \mathbb{N}$.

Suppose d_1 and d_2 are two gcds of a and b .

Since d_1 is a gcd of a and b , and $d_2 | a$ & $d_2 | b$,

$$d_2 | d_1$$

Similarly since d_2 is a gcd of a & b ,

$$d_1 | d_2$$

As $d_1, d_2 \in \mathbb{N}$, $d_1 = d_2$.

The gcd of a and b is denoted by
 $\gcd(a, b)$ or (a, b) .

Definition! Two natural numbers a and b are said
to be ~~Coprime~~ Coprime (or relatively prime) if

$$\gcd(a, b) = 1.$$

Ex: $\gcd(6, 7) = 1 \Rightarrow 6$ and 7 are Coprime.

Corollary: Two natural numbers a and b are
Coprime if and only if $ax+by=1$ for
some $x, y \in \mathbb{Z}$.

Proof: Let a and b are coprime.

$$\Rightarrow \gcd(a, b) = 1.$$

$$\Rightarrow \exists x \text{ and } y \text{ such that } ax+by=1.$$

Now suppose $\exists x, y \in \mathbb{Z}$ such that $ax+by=1$.

$$\text{Let } d = \gcd(a, b).$$

$$\Rightarrow d|a \text{ and } d|b.$$

$$\Rightarrow d|ax+by \Rightarrow d|1.$$

as $d \in \mathbb{N}$, we must have $d=1$.

Corollary: Let $a, b, c \in \mathbb{N}$. $\gcd(a, b) = 1$ &
 $a|bc$, then $a|c$.

Proof:

$$\gcd(a, b) = 1$$

$$\Rightarrow \exists x, y \in \mathbb{Z} \text{ s.t. } ax+by=1.$$

$$\Rightarrow acx+bcy=c.$$

$$a|ac \text{ and } a|bc$$

$$\Rightarrow a|acx+bcy \Rightarrow a|c.$$

Least Common Multiple (LCM): Let $a, b \in \mathbb{N}$. A positive integer l is called a least common multiple of a and b if it satisfies the following two conditions:

- ① $a|l$ and $b|l$.
- ② if $m \in \mathbb{N}$ is such that $a|m$ and $b|m$ then $l|m$.

Eg: lcm of 6 and 10 is 30.

Theorem: The least common multiple of two natural numbers always exist.

Proof: Let $a, b \in \mathbb{N}$.

Consider the set $L = \{m \in \mathbb{N} \mid a|m \text{ and } b|m\}$.

Clearly $L \subseteq \mathbb{N}$.

$L \neq \emptyset$ as $ab \in L$.

By well-ordering principle, L has a smallest element, say l .

Let $m \in \mathbb{N}$ be such that $a|m$ and $b|m$.

By division algorithm $\exists q$ and r such that

$$m = lq + r \quad \text{where } 0 \leq r < l.$$

If $r \neq 0$, $r = m - lq$ $a|m$ and $a|l \Rightarrow a|m - lq = r$

Similarly $b \mid r$.

This contradicts the fact that l is the least element in L .

$$\Rightarrow r = 0.$$

hence $l \mid m$.

So l is the lcm of a and b .

Remark: Least common multiple is unique.

Proof: Let $a, b \in \mathbb{N}$.

Let l_1 and l_2 are least common multiple of a and b .

As l_1 is lcm of a & b and $a \mid l_2, b \mid l_2$

$$l_1 \mid l_2$$

Similarly $l_2 \mid l_1$

As $l_1, l_2 \in \mathbb{N}$, we must have $l_1 = l_2$.

We will denote the lcm of a & b by

$$\text{lcm}[a, b].$$